Muon Neutrino Charged Current Single Pion Cross Section on Argon using Run 1 NuMI Data

MicroBooNE Internal Note — Draft

Version 0.1

DRAFT STATUS — READ FIRST

This note mirrors the structure of the BNB CC1π internal note (`BNBInternal-NoteCC1pi_V0.91.pdf`, J. P. Detje, 4 February 2025) for the NuMI Run 1 FHC measurement.

Sections 1–5 describe the analysis *as it is actually configured in the code today*, and every number in them was read from the source or the configuration files. Sections 6–7 are *structural placeholders*. No cross-section values, efficiencies, purities, χ^2 values or figures appear anywhere in this note, because the analysis has not been re-run since the corrections described in Section 2.4. Nothing here should be quoted as a result.

Three items must be settled before this note can carry results at all; see Section 8.

Abstract

This note describes the extraction of the ν_μ charged-current single charged pion cross section on argon using MicroBooNE Run 1 FHC data from the NuMI beam. It is the NuMI counterpart of the BNB analysis of the same final state and is intended to be read alongside that note, which contains fuller descriptions of the selection variables and the boosted decision trees. The signal definition, binning, selection, uncertainty treatment and unfolding procedure are documented; the fake-data tests and results sections are reserved pending a complete re-run of the analysis chain.

Contents

1 Overview

The measurement is performed with the common `xsec_analyzer` framework: event selection via the `CC1mu1piXp` selection class, universe construction via `univmake`, and unfolding via the Wiener-SVD implementation in `CrossSectionExtractor`.

Section 2 describes the software and samples, Section 3 the signal definition and selection, Section 4 the uncertainty treatment, Section 5 the cross-section extraction and unfolding, Section 6 the fake-data tests and Section 7 the results.

The principal differences from the BNB analysis are summarised in Table 1.

	BNB analysis	**This analysis**
Beam	BNB	NuMI FHC
Runs	1–5	1 only
Flux systematic	`weight_flux_all`	`weight_ppfx_all` (PPFX)
Beamline geometry	not applicable	NuMI-specific; **not currently included**, see Section 4.6
Proton multiplicity	—	Xp (any number, no threshold)
Unfolding	Wiener-SVD	Wiener-SVD, D'Agostini as cross-check

Table 1: Principal differences from the BNB CC1π analysis.

2 Software and Samples

2.1 Framework

The analysis runs entirely within `xsec_analyzer`. The chain is:

1. `ProcessNTuples` — applies the `CC1mu1piXp` selection to raw PeLEE ntuples and writes `stv_tree` output (`run_process_ntuples.sh`).
2. `univmake` — builds systematic universe histograms from a bin configuration (`run_universe_maker.sh`).
3. `Unfolder` / `UnfolderNuMI` — extracts and unfolds the cross section (`run_unfolder.sh`).

2.2 Samples

Run 1 FHC, as configured in `configs/file_properties_numi.txt` (Table 2).

Sample type	**Purpose**
`onBNB`	beam-on data (see warning below)
`extBNB`	beam-off (EXT), 3 821 593 triggers
`numuMC`	NuMI overlay central value
`dirtMC`	dirt overlay
`detVarCV` + 8 variations	detector systematics

Table 2: Run 1 FHC sample set.

The `onBNB` slot currently holds NuWro fake data

The active entry is `xsec-ana-numi_nuwro_overlay_pion_ntuples_run1_fhc.root` (6.65×10^{20} POT). The real beam-on sample (`xsec-ana-neutrinoselection_filt_run1_beamon_beamgood.root`, 3.283×10^{20} POT, 7 809 962 triggers) is **commented out** at `configs/file_properties_numi.txt:31`.

The analysis is therefore configured as a *fake-data study*, not a real-data measurement. Section 7 cannot be filled until this is swapped back and the chain re-run.

2.3 Detector variation samples

Eight variations are used: `detVarLYdown`, `detVarLYrayl`, `detVarRecomb2`, `detVarSCE`, `detVarWMAngleXZ`, `detVarWMAngleYZ`, `detVarWMX` and `detVarWMYZ`.

The BNB note additionally lists `detVarLYatten`; it is not present in the NuMI sample set here. As in the BNB analysis, the WireMod dE/dx variation is not used, in favour of the recombination variation.

2.4 Corrections applied since the previous iteration

The following were found and fixed during a code review of this analysis. All of them affect the numbers a run would produce, which is why no results predating them should be carried forward.

- **Muon momentum estimator.** The estimator combining range (contained) and MCS (uncontained) was computed but never written to the output tree, so the unfolding binned on MCS alone. It is now branched as `candidate_muon_mom_reco` and `ccpi_pmu_bin_config.txt` uses it.
- **Pion momentum estimator.** Reconstructed pion momentum was a proton-hypothesis range *kinetic energy* binned against a true *momentum* on identical bin edges. It is now the muon-hypothesis range momentum.
- **Per-category systematics.** Six `MCFullCorrCategory` entries evaluated to identically zero and have been removed; see Section 4.6.
- **Background true bins.** `MakeConfig` had stopped emitting them, so any regenerated bin configuration would have had no background subtraction at all.
- **Diagnostic histograms** are now filled with the central-value weight (`spline_weight_` $\times$ `tuned_cv_weight_`).

3 Signal and Selection

3.1 Signal definition

An event is signal if, in truth:

- the neutrino vertex lies inside the fiducial volume, $10 < x < 246$ cm, $-101 < y < 101$ cm, $10 < z < 986$ cm;
- the interaction is charged-current;

- the neutrino is ν_μ (`|mc_nu_pdg| == 14`, see Section 8);
- there is exactly one muon above threshold;
- there is exactly one charged pion;
- there are no neutral pions, no kaons and no heavier mesons;
- any number of protons is allowed, with no momentum threshold — the "Xp" in `CC1mu1piXp`;

and the event falls within the measured phase space:

$$p_\mu > 0.15 \text{ GeV}/c, \qquad p_\pi > 0.175 \text{ GeV}/c, \qquad \theta_{\mu\pi} < 2.6 \text{ rad}. \tag{1}$$

The phase-space cuts are applied identically in `define_signal()` and `categorize_event()`, so signal and category assignment cannot disagree.

3.2 Binning

Five differential observables are measured; bin edges as configured are given in Table 3.

Observable	Config	true/reco	Edges
p_μ	`ccpi_pmu`	23/22	0.15, 0.20, 0.30, ..., 2.00, 2.30, 2.60, 3.00 GeV/c
p_π	`ccpi_ppi`	6/5	0.175, 0.20, 0.30, 0.40, 0.50, 1.00 GeV/c
$\cos\theta_\mu$	`ccpi_costhmu`	13/12	−1.0, −0.5, 0.0, 0.2, 0.4, 0.55, 0.65, 0.75, 0.82, 0.88, 0.93, 0.97, 1.0
$\cos\theta_\pi$	`ccpi_costhpi`	13/12	−1.0, −0.7, −0.4, −0.2, 0.0, 0.2, 0.35, 0.5, 0.65, 0.78, 0.88, 0.95, 1.0
$\theta_{\mu\pi}$	`ccpi_thmupi`	10/9	0.0, ..., 2.513, 2.600 rad

Table 3: Binning for the five differential observables.

Each true-bin count is one greater than the corresponding reco count: the extra true bin is the background bin, defined by inverting the signal definition (`!CC1mu1piXp_MC_Signal`). All configurations are single-block with no sideband reco bins.

3.3 Particle identification

Three BDTs are used, trained separately and applied via `TMVA::Reader` (Table 4).

Reader	Weights	Role
`tmvaReader`	`dataset_MIP_BDT_no_len`	MIP identification
`tmvaReader_mu`	`dataset_muon_BDT`	muon — **booked but never evaluated**, see Section 8
`tmvaReader_pi`	`dataset_pion_BDT_no_len`	pion

Table 4: Boosted decision trees used for particle identification. Weight files live under `booster_decision_tree/`.

Input variables are the three-plane Bragg likelihood ratios (`trk_bragg_p_v`, `trk_bragg_mu_v`, `trk_bragg_mip_v`), the LLR PID score, the track score, the track length, and the SCE-corrected track end position. The pion reader omits the track length.

3.4 Event selection cuts

Applied in order; an event passes only if all are satisfied:

1. software trigger
2. reconstructed vertex in the fiducial volume
3. topological score
4. muon candidate is track-like
5. pion candidate is contained
6. muon candidate not in a detector gap
7. pion candidate not in a detector gap
8. shower cut
9. muon–pion opening angle cut
10. final track-multiplicity cut

3.5 Selection efficiency and purity

To be filled from a complete `ProcessNTuples` pass. `finalize()` prints the efficiency and purity; these are now central-value weighted (Section 2.4) and therefore differ from any previously recorded values.

4 Uncertainties

The uncertainty budget is configured in `configs/ccpi_systcalc_numi.conf`. The total is

$$\text{total} = \text{detVar} + \text{flux} + \text{reint} + \text{xsec} + \text{POT} + \text{numTargets} + \text{SimStats} + \text{DataStats}. \tag{2}$$

4.1 Flux

PPFX hadron-production multisims via `weight_ppfx_all`. A 2 % fully-correlated normalisation uncertainty is applied for POT (`POT MCFullCorr 0.02`).

4.2 Interaction

GENIE multisims (`weight_All_UBGenie`) plus nine unisim knobs (`AxFFCCQEshape`, `DecayAngMEC`, `NormCCCOH`, `NormNCCOH`, `RPA_CCQE`, `ThetaDelta2NRad`, `Theta_Delta2Npi`, `VecFFCCQEshape`, `XSecShape_CCMEC`) and the two second-class-current form factors (`xsr_scc_Fa3_SCC`, `xsr_scc_Fv3_SCC`), matching the BNB treatment.

4.3 Reinteraction

`weight_reint_all` multisims.

4.4 Detector

The eight detector variation samples of Section 2.3, evaluated as type `DV`.

4.5 Target

A 1 % normalisation uncertainty on the number of argon nuclei in the fiducial volume (`numTargets MCFullCorr 0.01`), as in the BNB analysis.

4.6 Known gaps in the uncertainty budget

Two components are **not** included, and the total above is correspondingly under-estimated.

NuMI beamline geometry. The beamline variation weights (`weight_Horn_2kA`, `weight_Horn1_x_3mm`, `weight_Beam_spot_*` and others) are absent from the processed ntuples. `instructions_numi.txt` documents `AddBeamlineGeometryWeights` as the tool that injects them, but no driver script invokes it, and the required histogram file (`NuMI_Geometry_Weights_Histograms.root`, canonical copy at `/exp/uboone/data/users/pgreen/NuMIFlux/NewFluxFiles/`) is not available locally. This systematic has no BNB counterpart and must be restored before the measurement is complete.

Background normalisation by category, and dirt normalisation. Six `MCFullCorrCategory` entries were removed because they evaluated to identically zero: they matched true bins by exact string equality against `"category == N"` while the bin configurations declare a single `"!CC1mu1piXp_MC_Signal"` background bin. Their category indices were additionally inherited from the NuMI CC1e analysis and did not correspond to this selection's categories. Removing them changed no number — verified with 152 histograms bit-identical. Full detail is recorded in `configs/ccpi_systcalc_numi.README.md`.

For reference, the BNB uncertainty breakdown comprises `EXTstats`, `MCstats`, `POT`, `detVar_total`, `flux`, `numTargets`, `reint` and `xsec_total` — the same set retained here, with no per-category or dirt term.

5 Cross-section Extraction

The flux-integrated differential cross section in truth bin α is obtained by subtracting the background from the measured event rate, unfolding to truth space, and dividing by the integrated flux Φ, the number of argon targets T, and the bin width:

$$\left.\frac{d\sigma}{dx}\right|_{\alpha} = \frac{1}{T\,\Phi\,\Delta x_{\alpha}} \sum_{a} U_{\alpha a}\left(D_a - B_a\right), \tag{3}$$

where D_a is the measured event rate in reco bin a, B_a the estimated background, and $U_{\alpha a}$ the unfolding matrix, which absorbs both the reco-to-truth mapping and the efficiency.

Targets are counted as *argon nuclei* in the fiducial volume, so the result is per-nucleus rather than per-nucleon, matching the BNB convention ($T = 1.03 \times 10^{30}$ nuclei there). For NuMI the fiducial volume additionally excludes the dead region $675.1 < z < 775.1$ cm.

Results are quoted in units of 10^{-38} cm^2/Ar, differential in the observable — so 10^{-38} cm^2/GeV/Ar for momenta and 10^{-38} cm^2/rad/Ar for angles.

Unresolved — see Section 8

The code currently divides by 10^{39} in NuMI mode and 10^{38} otherwise, and `Unfolder.C` does not divide by the bin width while `UnfolderNuMI.C` does. Both must be settled before any number in Section 7 is meaningful.

5.1 Unfolding

Wiener-SVD with the filter enabled and second-derivative regularisation (`Unfold WienerSVD 1 second-deriv`), chosen over D'Agostini after a variant study: it suppresses the bin-to-bin oscillation arising from sparse Run 1 statistics. D'Agostini with four iterations remains available as a cross-check via `configs/ccpi_xsec_config_numi_range_dagost.txt`.

As in the BNB analysis, the unfolded result must be compared to predictions smeared by the additional smearing matrix A_C.

6 Fake-Data Tests

Structure only — no results.

6.1 GENIE closure test

Treat the GENIE central value as data, add the proportionally scaled beam-off sample, unfold, and confirm that the GENIE CV truth distribution is recovered. Driven by `configs/ccpi_xsec_config_numi.txt`.

6.2 NuWro fake-data study

The same procedure with the NuWro overlay (`configs/ccpi_xsec_config_numi_nuwro.txt`, driven by `run_nuwro_observables.sh` across all five observables). As in the BNB analysis, only statistical and model uncertainties apply, since the samples differ only in the generator model.

Per-observable results to be added: reco-space selection histograms, reco- and truth-space bin correlations, confusion matrices, unfolded cross sections and additional smearing matrices, for each of $\cos\theta_\mu$, $\cos\theta_\pi$, p_μ, p_π *and* $\theta_{\mu\pi}$.

7 Results

Empty pending a real-data run. This requires, in order: the real beam-on sample restored in `file_properties_numi.txt`; the full chain re-run from `ProcessNTuples`, since the corrected momentum branches do not exist in the current processed files; and Section 8, item 1 resolved.

This section should contain, per observable and for the total, the unfolded cross section with the full uncertainty breakdown, and generator comparisons.

8 Open Items

1. **Cross-section units and bin-width normalisation.** `conversion_factor()` divides by 10^{39} in NuMI mode against 10^{38} for BNB, while the BNB note quotes 10^{-38} cm^2/Ar — a factor of ten discrepancy in convention. Separately, `UnfolderNuMI.C` divides by the bin width and `Unfolder.C` does not, so the two binaries disagree on the same bin. The BNB note is unambiguous that the result is differential and per-Ar.

2. **Beamline geometry systematic.** See Section 4.6.

3. **Anti-neutrino contamination.** The signal definition uses `|mc_nu_pdg| == 14`, admitting $\bar{\nu}_\mu$ charged-current events into both the signal and the efficiency denominator. All three sibling selections in the framework use an exact comparison. If the flux normalisation is ν_μ-only, this biases the result.

4. **Unused muon BDT.** `tmvaReader_mu` is constructed and booked but never evaluated, and its input variables are never filled, suggesting that an intended muon-PID cut was dropped.

Prepared as a companion to `BNBInternalNoteCC1pi_V0.91.pdf`.
Analysis code state: see git history on branch `fix/systematics-warning`.